

Consumer Acceptance of Virtual Influencers on Vietnamese Social Media — Analysis Playbook

How to turn the crawled Facebook comments into publishable marketing evidence: what “acceptance” means and how to measure it, how to code thousands of Vietnamese comments with an AI assistant without losing academic credibility, which analysis produces the Normalization Crossover Point, what the figures should say to a business reader, and which references to cite.

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1. The business question and the one figure the paper stands on

Brands in Vietnam are already using AI-generated personas. The managerial problem is timing and risk: *when does an audience stop talking about a virtual influencer as a “fake thing” and start talking about her as a persona they relate to – and does commercial receptivity follow?* The paper's contribution is a measurable, time-stamped answer: the **Normalization Crossover Point (τ^*)**, the week in which relational (parasocial) talk durably overtakes uncanny/artificiality talk in public discourse, and whether product-related talk rises after it.

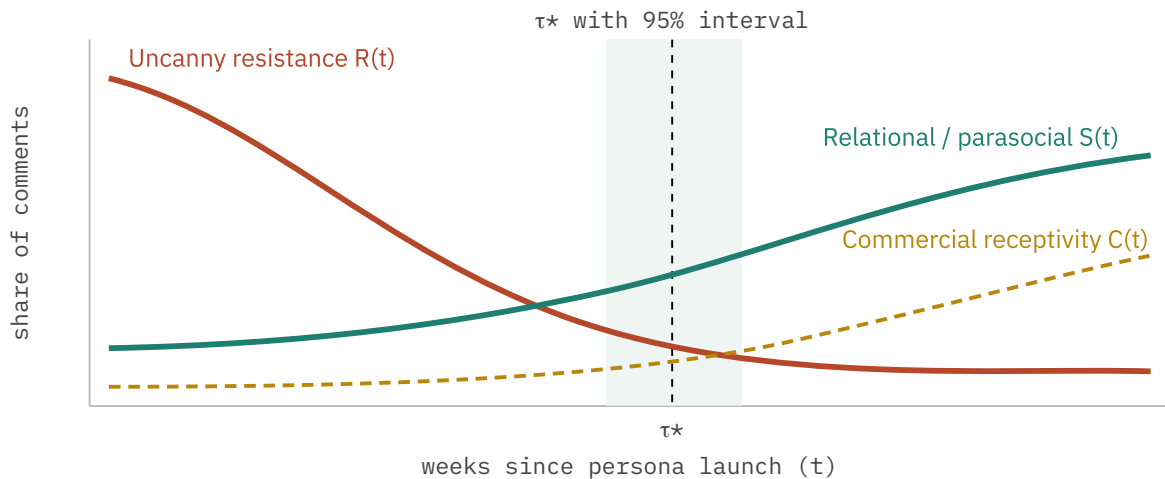


Figure 1 (target). Three audience responses over a persona's life on the platform. The paper lives or dies on whether these curves are measured credibly — not on a sentiment pie chart.

Everything below is in service of producing Figure 1 credibly for a marketing reviewer: a transparent sample, constructs grounded in consumer theory (uncanny valley, mere exposure, parasocial interaction, source credibility), coding validated by Vietnamese human coders, and statistics that state uncertainty honestly.

2. Data collection — documented for the Methodology section

This section is written so it can be lifted, trimmed, and pasted into 3.1 *Research design* and 3.2 *Instrumentation*. Numbers refer to the corpus as of 2026-08-21.

2.1 Research setting and sampling frame

- **Platform.** Facebook — the dominant social platform in Vietnam and the channel on which Vietnamese virtual influencers (VIs) publish and receive public comments. Study unit hierarchy: persona → post/reel → public comment/reply.
- **Personas.** Vietnamese-language, persistently active VI pages. Collected so far: *Tiểu Mỹ* (facebook.com/tieumyday), an AI-generated singer/persona with both organic (daily-life, music) and product-oriented (e.g., “*Lần đầu bán quạt...*”) posts. Target: 4–6 personas with different levels of human-likeness and different launch dates, so that curves can be compared across personas and the result is not one persona's story.
- **Observation window.** All public posts from page launch (time zero, $t = 0$) to collection date. For *Tiểu Mỹ*: 2026-03-10 → 2026-08-19 (23 weeks).
- **Unit of analysis.** The public comment (top-level or reply). Unit of time: week since launch (2-week bins as sensitivity).
- **Inclusion.** Vietnamese-language public comments on the persona's own posts/reels. Exclusion: comments posted by the persona account itself, spam/bot content, non-Vietnamese text, platform alt-text artefacts.

2.2 Collection instrument

Comments were collected with a purpose-built browser-automation crawler (Python, Playwright on Chromium) that reproduces what a logged-in user sees on the public page — it does not use private APIs or access non-public content. The procedure per persona:

01 Timeline discovery. Scroll the page feed and the Reels tab; record every post/reel with its feed position, caption, publication date, reactions, shares and Facebook's displayed comment count (used later as a coverage denominator).

02 Post opening. Open each post (preferably by permalink — this path returns full, absolute timestamps and stable author identifiers in 97–100% of cases, versus 15–

34% via the feed lightbox).

03 Thread expansion. Switch the comment sort to “All comments”, repeatedly expand “View more comments / replies” up to a per-post cap (600 expansion rounds or 15 minutes), and read comment text, author, author id, timestamp, parent relation and content type from the rendered DOM.

04 Persistence & resumability. Each post is written to SQLite (`data/db/scrapper.sqlite`) as soon as it is done; runs can be resumed and are merged by union, so repeated runs deepen coverage rather than overwrite.

05 Coverage measurement. For each post, collected comments are compared to the displayed comment badge (`ui_comment_count`); the gap is stored per post and used in robustness analyses.

2.3 Corpus collected so far (after de-duplication)

5,888

distinct comments

265

posts/reels with
comments (474
discovered)

23

weeks (2026-W12 →
W34)

3,783

unique commenters

346

commenters with ≥ 3
comments (33% of
all comments)

$\approx 1,960$

replies addressed to
another user

CHARACTERISTIC	VALUE	IMPLICATION FOR ANALYSIS
Content type	photo 2,382 · post 680 · reel 751 · unknown 2,075	Code content type per post; reels partially collected (≈ 56 remaining).
Timestamp precision	1,885 absolute · 3,247 “N weeks ago” · 756 “N days/hours ago”	Relative stamps resolved against run time (± 1 week). Weekly bins acceptable; 2-week bins as sensitivity.
Weekly volume	min 3 (W28) · max 2,278 (W33)	Volume is driven by viral posts; analysis must control for post (clustering), not use raw weekly shares.
Concentration	one viral photo post = 1,357 comments (23%)	Rerun every model excluding this post.
Coverage	$\approx 100\%$ on posts with ≤ 30 comments; 50–60% on posts with > 400	Report coverage; rerun with posts $\geq 50\%$ coverage only.
Comment length	median 33 characters; 4.5% ≤ 5 characters	Short, slang-heavy, emoji-rich register (teencode) — code in raw Vietnamese, never via machine translation.
Repeat commenters	346 authors ≥ 3 comments, 33% of comments	Enables a within-commenter robustness subsample (Layer B).

Commercial stimulus is almost absent for *Tiểu Mỹ*. Of 477 posts, ~17 contain commercial vocabulary and most of those are self-promotion (YouTube channel, MV, Zalo). Only a handful are product/selling posts (“*Lần đầu bán quạt...*”, “*Mời cả nhà ghé mua ‘quả thông minh’*”) plus one affiliate link posted by the persona in the comments. There is no third-party brand partnership. Consequences: (1) code a 4-level post variable `commercial_post` $\in \{\text{none, self_promo, own_product/affiliate, brand_sponsored}\}$ instead of a binary “sponsored”; (2) for this persona RQ4 can only be tested as *spontaneous* commercial receptivity (unprompted asks about what she wears/uses) and must be labelled exploratory; (3) at least two of the remaining personas must have real brand-tied posts (Facebook “Paid partnership” label, #ad, tagged brands) if H3/H4 are to be tested properly; (4) seller spam in comments (“*Mua tại đây* 📌”) is `spam=1`, never commercial receptivity, and “*xin link nhạc*” is content interest, not commercial.

2.4 Data cleaning rules (pre-registered in the paper)

- Remove exact duplicates (same post + author + text) and comments by the persona's own account (35).
- Remove Facebook alt-text artefacts captured as text (“*May be an image of...*”, “*Avatar is pointing...*”).
- Flag (not delete) emoji-only comments — they count toward engagement volume but carry no construct label.
- Flag suspected spam: single-word “Follow”, identical text repeated by one author across posts, links.
- Flag non-Vietnamese text with a language detector set permissively (teencode confuses detectors); exclude in a robustness run.
- Hash author identifiers before analysis; keep only the hash for repeat-commenter analysis.

2.5 Ethics and limitations statement (draft wording)

Data consist of publicly visible comments on public brand/persona pages, collected for non-commercial academic research; no private messages, friend-only content, or profile data were accessed. Author identifiers were pseudonymised by one-way hashing before analysis and no individual is identified in reporting. Collection reproduces an ordinary logged-in browsing session and respects platform rate limits; comments on highly viral posts are under-sampled (50–60% coverage), which we address through coverage-based

robustness checks. Timestamps shown by the platform as relative (“N weeks ago”) were converted to calendar weeks with ± 1 -week precision.

Mention, don't hide: automated collection from Facebook sits in a grey zone of the platform's terms even when only public content is read. Marketing journals generally accept it with a clear ethics statement and pseudonymisation; say it explicitly in Limitations rather than leaving a reviewer to raise it.

3. What we measure: from “sentiment” to four consumer responses

A sentiment score answers “is the audience happy?”. Your research questions ask something more specific and more useful to managers: *what kind of relationship does the audience have with the persona, and is it moving toward commercial readiness?* Two observations from the corpus show why sentiment alone misleads:

- “Con nhỏ này hôm nay lại đi quảng cáo nữa à =)))” — mildly negative, yet the commenter treats the VI exactly like any human influencer. That is **normalization without positivity**.
- “biết rằng tạo bằng AI nhưng ngoài thực tìm ra được em này không ta” — artificiality acknowledged, yet the response is attraction and curiosity. Positive sentiment, still artificiality-focused.

So the primary measurement is **four consumer-response constructs**, each coded present/absent, several allowed per comment. Sentiment is kept as a secondary, descriptive variable.

CONSTRUCT (PAPER NAME)	WHAT IT CAPTURES (THEORY)	OBSERVABLE IN COMMENTS	EXAMPLES FROM THE CORPUS
Uncanny resistance	Perceptual-mismatch / uncanny-valley discomfort with a near-human artificial entity (Mori; Kätsyri et al.; MacDorman & Chattopadhyay)	Focus on fake/strange appearance, eyes, voice, motion; eeriness; rejection <i>because</i> artificial	<i>“AI hát ghép nên nghe k thật” · “mỗi lúc 1 khác chả thấy đồng nhất” · “T thà xem phim hoạt hình”</i>
AI skepticism	Technology distrust / deception concern — a separate negative mechanism, <i>not</i> uncanny valley	Anti-AI opinions, “AI trash”, brands using AI are untrustworthy	<i>“khổ thân thế hệ hiện giờ, sống chung với rác AI”</i>
Parasocial / relational orientation	Treating the persona as a social actor (Horton & Wohl; CASA — Nass & Moon; Gambino et al.)	Direct address (<i>chị/em/bé/bà</i>), affection, missing her, personality attribution, asking her to post/reply, defending her, jokes <i>with</i> her	<i>“Sắp thuộc rồi chú Khoa ơi” · “nói AI bé nó buồn đấy” · “Ra video tiếp đi”</i>
Commercial receptivity	Behavioural textual signals consistent with commercial interest (source credibility / influencer persuasion —	Asks link, price, brand, where to buy; wants to try/buy; welcomes a collab	<i>“Quạt, quạt gì cơ” (curiosity) · “xin link” · “Collab này hợp quá”</i>

Sentiment (secondary)	Valence	negative / neutral / positive	—
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Naming for the paper. Call the downstream construct *commercial receptivity / commercial intent signal*, not purchase intention — purchase intention is a psychometric construct measured by survey (e.g., Dabiran et al., 2024), and reviewers will object if a comment is labelled as intention. Call the phenomenon *market-level discourse normalization*, because different people comment in different weeks; the repeat-commenter subsample is the only within-person evidence.

4. Coding at scale: AI-assisted content analysis with human validation

Marketing journals now accept large language models as coders *provided* the workflow is human-led and validated on held-out human-coded data (Li et al., 2024, *Marketing Science*; Arora et al., 2025, *Journal of Marketing*; Carlson & Burbano, 2026, *SMJ*; Törnberg, 2024). This is standard content analysis (Krippendorff, 2018) with the LLM replacing the third, fourth... coder. For Vietnamese specifically, zero-shot LLM sentiment accuracy is only moderate (GPT-4 F1 $\approx .67$ on Vietnamese benchmarks — Truong et al., 2024), which is exactly why the human gold standard is the centre of the method, not a formality.

You have chosen **DeepSeek V4 Flash**: open-weight, cheap (the whole corpus of 5 personas costs well under US\$5), and version-pinnable — which also answers the reproducibility concern raised about closed models (Ollion et al., 2024).

4.1 The workflow (write this as Section 3.2 Instrumentation)

- 01 **Clean & time-resolve** the corpus (Section 2.4); attach to each comment the post caption, content type, sponsorship flag, and the parent comment if it is a reply.
- 02 **Pilot coding (humans, ~200 comments)**. Stratified by persona $\times$ month $\times$ content type, oversampling comments that contain resistance or product vocabulary so the rare categories appear. Output: the codebook — definition, include/exclude rules, 3–5 raw Vietnamese examples per code, and documented emergent themes (e.g., joking anthropomorphism, ironic “she’s advertising again” talk).
- 03 **Gold standard (two Vietnamese coders, 600–800 comments)**. Independent coding on raw Vietnamese; disagreements adjudicated. Report Krippendorff’s α per construct ($\geq .67$ acceptable, $\geq .80$ good) and Cohen’s κ .
- 04 **Freeze; split gold into development ($\approx 40\%$) and test ($\approx 60\%$)**. The prompt is tuned on development only; the test set is touched once.
- 05 **LLM coding**. One comment per request (or ≤ 25 per batch with ids echoed), temperature 0, JSON output; record model id, date, prompt, parameters. Re-code a

5% sample to report the model's self-consistency.

- 06 **Evaluation on test.** Per construct: precision, recall, F1, confusion matrix; macro-F1 overall. Target recall $\geq .70$ on the rare constructs. Never report accuracy alone — with 97% of comments not commercial, “predict none” scores 97%.
- 07 **Code the full corpus** with the frozen prompt. Then carry coding uncertainty forward: (a) repeat the main analysis on the human-coded subset and show the same direction; (b) bootstrap by post; (c) optionally apply the design-based correction of Egami et al. (2023).

VALIDATION TABLE TO PUT IN THE PAPER	LEVEL	DEFENSIBLE VALUE
Krippendorff's α (coder 1 vs coder 2)	per construct	$\geq .67$ / $\geq .80$
Precision · Recall · F1 (LLM vs gold, test set only)	per construct	recall $\geq .70$ on rare classes
Macro-F1	overall	report
LLM self-agreement on repeat run	5% sample	$\geq .90$
Direction consistency	human subset vs full corpus	same sign of trends

The full codebook prompt is in Appendix A. It is multi-label (a comment can be resistance + relational + commercial at once, e.g. “*Nhìn mặt vẫn hơi giả nhưng bà này nói chuyện hài ghê, cái áo đẹp xin link*”), which is what later lets you show *coexistence* — a finding in itself.

5. Aspect-level reading: what consumers reject and what they embrace

To give managers actionable detail, add an **aspect-based sentiment** layer: for each comment, which *aspect* of the VI is being judged and with what valence. Use a fixed set of aspect categories (the approach used in Vietnamese ABSA benchmarks — UIT-ViSFD, VLSP-2018 — and in SemEval), not free-text aspect extraction, which breaks on teencode.

ASPECT	COVERS	MANAGERIAL READING
APPEARANCE	face, eyes, skin, body, visual realism	Design/rendering quality; the classic uncanny trigger
VOICE_MOTION	voice, lip-sync, singing, movement	Production quality; often the last mismatch to disappear
AI_NATURE	being AI per se, AI in society	Disclosure/transparency strategy; ethical positioning
PERSONA	personality, humour, charm, relationship talk	Character strategy — this is where relational value is built
CONTENT	the song/video/story itself	Content strategy (control)
PRODUCT_BRAND	product, price, link, collab, ad	Commercial readiness and fit of the sponsorship

Each aspect receives negative / neutral / positive. The insight is the **aspect × week net-polarity heatmap**: managers see appearance complaints fading, persona praise rising, product curiosity appearing — and can read *which* lever (rendering, character, disclosure, product fit) is moving. It also yields a second, independent crossover estimate (the week `PERSONA:positive` overtakes `APPEARANCE:negative + VOICE_MOTION:negative`); if it falls inside the interval of the construct-based τ^* , you have convergent evidence.

6. Analysis: trajectories, crossover, and commercial receptivity

Written for a business reader first; model specifications are in Appendix B. Tools: Python for cleaning and LLM coding; R (`mgcv` , `segmented` , `boot`) for modelling.

6.1 Step 1 — Trajectories (RQ1, RQ2; H1, H2)

For each construct, estimate the *probability that a comment posted in week t carries that construct*, with a flexible curve over time rather than a straight line (mere-exposure effects are often non-linear — Montoya et al., 2017), while controlling for what the persona posted (sponsored vs organic, content type, visual realism, AI disclosure) and for the fact that comments cluster inside posts and personas. The method is a generalized additive model with random effects (Wood, 2017; Pedersen et al., 2019): it produces smooth curves with confidence bands, and the *rate of change* (resistance decay rate, socialization growth rate) comes directly from the slope of those curves (Simpson, 2018).

Why the controls matter to the story: if the VI's graphics, voice and storytelling improve over six months, negative talk can fall because the product got better, not because the audience normalized. Coding realism and sponsorship per post is what lets you attribute the change to audience adaptation.

6.2 Step 2 — The crossover point (RQ3)

Compute the difference curve $D(t) = S(t) - R(t)$. Define τ^* *before looking at the result*: the earliest week at which the lower 95% bound of $D(t)$ is above zero for at least three consecutive weeks (a pre-registered decision rule, so one noisy week cannot create a “crossover”). Quantify uncertainty by re-sampling posts (cluster bootstrap, 500–1,000 replications; Cameron et al., 2008; Lee et al., 2015) and re-estimating τ^* each time. Report “ $\tau^* \approx$ week 14, 95% CI [12, 18]” — or “no sustained crossover within the 23-week window; resistance fell from .18 to .05 while relational talk rose from .12 to .21”. Both are publishable; do not force it.

6.3 Step 3 — Phase structure (robustness)

Your narrative proposes phases (visual resistance → social reframing → commercial accommodation). Test rather than assume it: segmented regression with data-estimated

breakpoints (Muggeo, 2003, 2008) tells you whether the weekly series supports zero, one or two breakpoints. If one, the paper reports two phases.

6.4 Step 4 — Commercial receptivity (RQ4; H3, H4)

Two versions of RQ4, depending on the persona. For personas *without* brand partnerships (e.g. *Tiểu Mỹ*): $C(t)$ measures *spontaneous* commercial receptivity — report as exploratory, drop the sponsored-share control (no variance), keep `commercial_post` level as a covariate. For personas *with* brand-tied posts: the full H3/H4 test applies. A persona with no commercial stimulus is also a *clean baseline* of normalization uncontaminated by advertising, which lets you answer “does sponsorship accelerate or delay normalization?” by comparing curves across personas.

Relate weekly commercial receptivity C_t to the previous week's balance D_{t-1} , controlling for how much sponsored/product content the persona posted that week and for last week's C . A positive, significant coefficient supports “a more favourable relational-to-resistance balance *precedes* more commercial-receptivity talk”. The before/after- τ^* comparison of C is descriptive; because τ^* is estimated from the same data it is not a causal interrupted-time-series design (Bernal et al., 2017) — say so.

6.5 Step 5 — Repeat commenters (Layer B)

346 commenters appear three or more times. Order their comments in time and tabulate first-label → last-label transitions (resistance → relational → commercial). It is a selective group, so present it as supporting within-person evidence, not the main result.

6.6 Robustness checks (commit to the list now)

RERUN	REASON
Weekly vs 2-week bins	“N weeks ago” precision
Full LLM-coded corpus vs human-coded subset	Coding validity
Exclude spam / non-Vietnamese	Bot and noise influence
Exclude the viral W33 post (1,357 comments)	Campaign shock
Each persona separately	One persona dominating
Sponsored vs organic posts separately	Commercial confound
Strict vs broad definition of resistance	Construct sensitivity
Crossover rule 2 / 3 / 4 consecutive weeks	Definition sensitivity
Posts with $\geq 50\%$ comment coverage only	Collection selection

6.7 Metric sheet — every indicator, its formula, input, tool, and the claim it supports

One input table (one row per comment: four 0/1 labels + sentiment + aspects + week + post + persona + controls). Three tools: Python/pandas (tables, descriptive indicators, P/R/F1), R `mgcv` + `gratia` + `boot` (curves, $D(t)$, τ^* , CI), R `segmented` (phases).

A. MEASUREMENT QUALITY (VALIDATION TABLE)

INDICATOR	FORMULA / DEFINITION	INPUT	TOOL	
Krippendorff's α	$\alpha = 1 - D_o/D_e$ (nominal)	2 Vietnamese coders $\times$ 600–800 comments $\times$ each construct	R <code>irr::kripp.alpha()</code> / <code>krippendorffsalpha</code>	
Cohen's κ (secondary)	$(p_o - p_e)/(1 - p_e)$	same	<code>irr::kappa2()</code>	
Precision / Recall / F1 per construct	$P = TP/(TP+FP)$; $R = TP/(TP+FN)$; $F1 = 2PR/(P+R)$	LLM labels vs gold, test set only	<code>sklearn.metrics.classification_report</code>	
Macro-F1	mean of class F1	same	<code>f1_score(average="macro")</code>	
Confusion matrix	TP/FP/FN/TN	same	<code>confusion_matrix</code>	
LLM self-agreement	% identical labels on a rerun	5% of corpus, temperature 0	Python	
Direction consistency	sign of dR/dt , dS/dt on human-coded subset vs full corpus	two GAM fits	R	

B. TRAJECTORIES (RQ1, RQ2 / H1, H2)

INDICATOR	FORMULA	INPUT	TOOL	REPORT
R(t), S(t), C(t)	P(label = 1 week t) from logit- GAM: logit P = $\beta_0 + s(t) +$ controls + $u_{\text{post}} + u_{\text{VI}}$	comment rows: uv/psi/cr , week , post_id , vi , sponsored , content_type , realism	<code>mgcv::bam(family=binomial)</code> → <code>predict(type="response",</code> <code>se.fit=TRUE)</code>	curves + 95% bands (Fig 1)
Resistance decay rate	$-dR/dt$ (finite- difference derivative of the smooth)	m_uv	<code>gratia::derivatives()</code>	weeks where < 0 → H1
Socialization growth rate	dS/dt	m_psi	<code>gratia::derivatives()</code>	weeks where > 0 → H2
Start–end levels	R(1) → R(T); S(1) → S(T)	predict at week 1, T	R	“R fell .18 → .05”
EDF, p of s(t)	non- linearity and significance of the time smooth	<code>summary(m)</code>	R	appendix table

C. CROSSOVER (RQ3)

INDICATOR	FORMULA	TOOL	REPORT
D(t)	$\hat{S}(t) - \hat{R}(t); SE \approx \sqrt{(SE_S^2 + SE_R^2)}$	from the two predicts (or inside the bootstrap)	Fig 2
τ^* (Normalization Crossover Point)	earliest week with lower 95% bound of $D(t) > 0$ for ≥ 3 consecutive weeks (pre-registered rule)	R loop over weekly grid	" τ^* = week 14" or "none within window"
95% CI of τ^*	cluster bootstrap by post, B = 500–1000: resample posts $\rightarrow$ refit m_{uv} , $m_{\psi} \rightarrow \tau_b^*$; percentiles 2.5/97.5	R <code>boot</code> or manual loop	" τ^* = 14.3, CI [12.1, 17.8]"
τ_k^* per persona	same, from <code>s(week, by=vi)</code>	same	Fig 3 strip
Number of breakpoints	AIC of 0/1/2 breakpoints on weekly series; <code>pscore.test</code>	<code>segmented::segmented(npsi=1,2)</code>	"data support k phases"
Rule sensitivity	τ^* under 2/3/4-week rule, 1/2-week bins	rerun	robustness table

D. COMMERCIAL RECEPTIVITY (RQ4 / H3, H4)

INDICATOR	FORMULA	INPUT	TOOL	REPORT
C_t (weekly)	$n(CR=1)/n$ in week t	weekly table	pandas	series
β_1 (H4, leading indicator)	$C_t = \alpha + \beta_1 D_{t-1} + \beta_2 C_{t-1} + \beta_3 \text{SponsoredShare}_t + \beta_4 \text{ProductShare}_t + \beta_5 \log(\text{volume}_t) + \varepsilon$	weekly table (persona fixed effects if pooled)	<code>glm(quasibinomial)</code> + <code>sandwich::NeweyWest</code>	β_1 , CI, p ; expect $\beta_1 > 0$
β_1, β_2 (H3)	$C_t = \beta_0 + \beta_1 S_t - \beta_2 R_t + \varepsilon$	weekly	glm	signs & CI
C before/after τ^*	mean $C(t < \tau^*)$ vs mean $C(t > \tau^*)$	weekly	descriptive	one sentence; not causal

E. INSIGHT INDICATORS (QUALITATIVE FINDINGS & MANAGERIAL IMPLICATIONS)

INDICATOR	FORMULA	TOOL	USED IN
Co-occurrence & lift	$P(A \wedge B)$; lift = $P(A \wedge B)/(P(A) \cdot P(B))$ for UV-PSI, PSI-CR, UV-CR	pandas crosstab	Tab 3 “ambivalence”
Normalization-positivity gap	share(PSI=1) – share(sentiment=pos), early vs late window	pandas	Fig 5
Net polarity by aspect × week	$(n_{\text{pos}} - n_{\text{neg}})/(n_{\text{pos}} + n_{\text{neg}} + n_{\text{neu}})$	pandas → ggplot heatmap	Fig 4
$\tau^*(\text{ABSA})$	week share(PERSONA:pos) > share(APPEARANCE:neg + VOICE_MOTION:neg), 3-week rule	as τ^*	convergent check
Repeat-commenter transition rate	among authors with ≥ 3 comments: % first label $\in$ UV and last $\in$ PSI/CR; transition matrix	pandas	Fig 6, Layer B
Distinctive vocabulary	log-odds ratio with Dirichlet prior, early vs late (Monroe et al. 2008)	Python	Fig 8
Coverage ratio	$n_{\text{collected}}/\text{ui_comment_count}$ per post	crawler	robustness filter $\geq .5$

F. CORPUS DESCRIPTION (TABLE 1)

Comments, posts, weeks, personas; comments/week (min–max); % replies; % emoji-only; % spam; % absolute vs relative timestamps; % comments on sponsored posts — all from the analysis table with pandas.

7. Telling the story: figures and tables for a marketing audience

A business reader should understand the finding from the figures alone. Keep one colour per response across the whole paper (resistance = brick, relational = teal, commercial = amber), put time on the x-axis everywhere, and avoid pies, word clouds and dual axes.

#	FIGURE / TABLE	WHAT THE READER TAKES AWAY
Fig 1	R(t), S(t), C(t) curves with confidence bands; τ^* shaded with its interval	“Resistance fades, relationship grows, here is when the balance tips.”
Fig 2	Difference curve D(t) crossing zero; the 3-week sustain window marked	“The tip is durable, not a blip.”
Fig 3	Small multiples per persona; strip of τ^* estimates with intervals	“Speed of normalization differs by persona (human-likeness, content mix).”
Fig 4	Aspect $\times$ week heatmap of net polarity (positive – negative)	“Appearance complaints fade first, voice later; persona praise leads; product curiosity follows.”
Fig 5	Slope chart: share of relational comments vs share of positive sentiment, early vs late window	“Normalization is not the same as liking.”
Fig 6	Alluvial/Sankey of repeat commenters' first $\rightarrow$ last response	“Individuals move in the same direction the market does.”
Fig 7	Commercial receptivity C(t) with τ^* marked + coefficient plot of D_{t-1} across specifications	“Commercial talk rises after the balance tips; robust across bins/lags.”
Tab 1	Corpus description (Section 2.3)	Transparent sample
Tab 2	Coding validation (α , P/R/F1, macro-F1)	Credible measurement
Tab 3	Co-occurrence of responses (resistance $\wedge$ relational, relational $\wedge$ commercial, ...)	“Ambivalence is common: people joke about the fake face and ask for the link in one breath.”
Tab 4	Exemplar comments per response $\times$ phase, raw Vietnamese + English gloss	Voice of the consumer; the qualitative findings section
Fig 8 (opt.)	Early-vs-late distinctive vocabulary (log-odds, Monroe et al., 2008)	“The words changed from <i>giả</i> , <i>AI</i> , <i>ghê</i> to <i>chị</i> , <i>xinh</i> , <i>link</i> .”

Produce figures in R/ggplot2 (with `gratia` for the smooth curves and `patchwork` for panels); export PDF/SVG at 85 mm or 178 mm width, fonts ≥ 7 pt. Use Python/Plotly only for exploration and the team dashboard.

8. What the paper can and cannot claim

YOU CAN SAY

- In Vietnamese public discourse, uncanny-related resistance toward VI X attenuated over N weeks while relational talk grew; the balance tipped at τ^* [CI] (or: did not tip within the window).
- A more favourable relational-to-resistance balance preceded higher commercial-receptivity talk (association, lagged).
- Normalization occurred without a matching rise in positive sentiment — audiences treat the VI as an ordinary influencer, including complaining about her ads.
- The crossover point is a practical *leading indicator* for timing promotional intensity, to be read with its interval.

YOU CANNOT SAY

- That individual consumers changed their minds (different people comment each week) — unless Layer B is strong, and then only as supporting evidence.
- That the crossover *causes* purchase intention or sales.
- That the uncanny valley is “temporary and self-decaying” in general — repeated exposure lowers eeriness, but liking depends on the persona's behaviour (Złotowski et al., 2015).
- Anything about Instagram/TikTok audiences — this is Facebook Vietnam.

Managerial implications you will be able to write: (1) a measurable readiness signal for when to shift a VI from awareness/character-building content to product integration; (2) which aspects (appearance, voice, persona, disclosure) drive resistance at each stage, i.e. where to invest; (3) evidence that character/relational content — not rendering quality alone — is what moves the audience toward commercial receptivity; (4) that AI disclosure and sponsorship density moderate the curves.

9. Roadmap to 28/08 — parallel workstreams, owners, and what to do now

Five workstreams run in parallel; the critical path is **crawl → pilot → gold set → AI coding → modelling → figures**. Anything in the Writing row can proceed without data. Colours:

data human coding AI coding modelling writing .

	Thu 21	Fri 22	Sat 23	Sun 24	Mon 25	Tue 26	Wed 27	Thu 28
Data & crawl	Crawl 2–4 more personas; set t=0 each Minh		Build analysis table Minh · App. C	Code posts: sponsored, disclosure, realism Minh/Trúc				
Human coding		Pilot 200 → code-book v1 TrúC+Minh	Gold set 600–800, two coders; α; freeze dev/test 2 VN coders					
AI coding			Annotate script; dry-run on pilot Minh · App. A	Tune on dev → one test eval → code corpus Minh				
Modelling				R scripts on pilot labels Minh · App. B	GAM curves, τ^* bootstrap, robustness; Figs 1–3 Minh	C(t) model, Layer B, ABSA; Figs 4–7 Minh		
Writing	Write 3.1–3.2 Method + ethics (from §2, §4) Minh		Lit review, framework, hypothesis wording TrúC		Results skeleton TrúC	Findings 4.1/4.2 Minh+Trúc	Discussion, limits, integrate All	Submit

Dependencies: gold set cannot start before codebook v1 (Fri evening); AI test evaluation cannot happen before gold is frozen (Sun); Figures 1–3 need the coded corpus (Mon). Writing rows never wait.

9.1 What to do now — the next 48 hours (Minh)

This is the advisor's plan in the PDF, applied to what the data audit showed. Each item names the concrete output.

-
- 01 Decide the persona list and t = 0 for each** (4–6 personas, varied human-likeness, Vietnamese audience; **at least two with real brand-tied posts** — check the page for Facebook’s “Paid partnership / Có trả phí hợp tác” label, #ad / #quangcao, tagged brands, product links — before crawling). Output: `data/personas.csv` with `vi_id`, `page_url`, `launch_date`, `human_likeness` (low/med/high), `notes`. Without `t = 0` you cannot build `week_since_launch` — the main predictor.
-
- 02 Run the crawl for the new personas** with the permalink/seed path where possible (97–100% absolute timestamps vs 15% on the feed path) and `caffeinate -dimsu`. Keep *Tiểu Mỹ* as is; ≈ 56 reels remain — optional.
-
- 03 Build the analysis table** (Appendix C). The logic already exists in this session's audit: dedup on post+author+text, resolve “N weeks ago” against run `started_at`, set `time_precision`, attach caption/parent, hash authors, compute `coverage_ratio`, strip alt-text artefacts. Output: `data/analysis/comments.parquet` + a one-page data report (counts per persona $\times$ week).
-
- 04 Draw the pilot sample (200)** stratified by persona $\times$ month $\times$ content type, oversampling comments matching the resistance / product keyword lists (keyword hit rates: resistance $\approx 1.6\%$, commercial $\approx 2.7\%$ — random sampling would give you ~ 3 and ~ 5 examples). Output: `data/coding/pilot_200.csv` with blank label columns; share with TrúC.
-
- 05 Write `tools/annotate_deepseek.py`** from Appendix A: JSON-only, temperature 0, id echo, retries, logs model id + date; dry-run on the pilot. Output: cost and a first confusion look against your own pilot labels (not the test set).
-
- 06 Draft Section 3.1–3.2 of the paper** from §2 and §4 of this playbook (sampling frame, instrument, cleaning rules, ethics, validation design). This is the Methodology deliverable due 15/08 that is now late — it needs only what you already have.
-

9.2 In parallel (TrúC)

- Brief the two Vietnamese coders; agree decision rules for the hard cases in the codebook (“*AI hả?*” is not resistance; “*lại quảng cáo nữa*” is relational, not commercial; teasing counts as relational).

- Finalize hypothesis wording with the construct names from §3 (“commercial receptivity discourse”, “market-level discourse normalization”) and the theory mapping (uncanny valley / perceptual mismatch → resistance; mere exposure → time; PSI/CASA → relational; source credibility → commercial).
- Pre-register, in the methods text, the crossover rule (lower 95% bound of $D(t) > 0$ for 3 consecutive weeks) and the robustness list in §6.6 — before results exist.

9.3 Decisions to lock before modelling

DECISION	RECOMMENDATION	WHY
Time zero	First public post of the persona page	Observable, same rule for all personas; “market entry” is ambiguous
Time bin	Week; 2-week as sensitivity	3,247 stamps have ± 1 -week precision
Gold-set size	600–800 (≥ 400 minimum)	Rare classes need ≥ 40 –60 positives in test for usable recall estimates
Crossover rule	3 consecutive weeks, lower CI > 0	Advisor's example; report 2 and 4 as sensitivity
Outcome naming	Commercial receptivity / intent signal	Avoid “purchase intention” without a survey
Unit of claim	Market-level discourse normalization	Different commenters each week; Layer B only supports

If time runs out, this is the cut line. Keep: cleaned corpus for ≥ 3 personas; gold set ≥ 400 with α ; DeepSeek validation on test; GAM curves + $D(t)$ + bootstrap τ^* with the 3-week rule; 2-week-bin and human-subset robustness. Defer (and say so): segmented-regression phase test, ABSA heatmap, DSL correction, lagged commercial model.

10. Appendices

Appendix A — Codebook prompt for DeepSeek V4 Flash (multi-label, JSON)

SYSTEM

You annotate Vietnamese Facebook comments about a virtual (AI-generated) influencer for an academic marketing study. Judge ONLY the comment text; use caption/parent for context. Labels are independent; several can be 1. Answer with JSON only.

CODEBOOK

uv_resistance (0/1): the comment's focus is the persona's artificial/synthetic appearance, eeriness, or perceptual mismatch (face, eyes, voice, motion; "giả", "đơ", "ghê", "nhìn sai sai", "không phải người thật", "AI mà ..."), or rejection because it is fake. Exclude general anti-AI opinions (→ ai_skepticism) and neutral questions such as "AI hả?" (→ uv_resistance=0, ai_mention=1).

ai_skepticism (0/1): distrust of AI/technology as such, deception, job loss, "rác AI", brands using AI are untrustworthy. Not about how she looks.

parasocial_orientation (0/1): treats the persona as a social actor: direct address (chị/em/bé/bà + verb), affection, missing her, personality attribution ("bà này hài"), asking her to reply/post more, defending her, joking WITH her, relational continuity ("hôm qua chị..."). Include even if the tone is teasing or negative ("lại đi quảng cáo nữa à").

commercial_acceptance (0/1): product/brand curiosity or intent: asks link, price, where to buy, brand name of something she wears/uses/sells; wants to try/buy; welcomes a collab/ad. Exclude complaints that it is "quảng cáo"; exclude requests for the SONG/VIDEO link ("xin link nhạc" → CONTENT interest, not commercial); exclude seller advertisements posted by other users (→ spam=1).

sentiment: "neg" | "neu" | "pos"

ai_mention (0/1): mentions AI/ảo/giả at all.

spam (0/1): bot, advert, copy-paste, unrelated to the persona.

aspects: list of {"cat":

APPEARANCE|VOICE_MOTION|AI_NATURE|PERSONA|CONTENT|PRODUCT_BRAND,

"pol": neg|neu|pos}; empty list if none.

INPUT

```
{"id": 8812, "persona": "tieumy", "caption": "...", "parent": "...",  
  "text": "Nhìn mặt vẫn hơi giả nhưng bà này nói chuyện hài ghê, cái áo đẹp xin link"}
```

OUTPUT

```
{"id": 8812, "uv_resistance": 1, "ai_skepticism": 0, "parasocial_orientation": 1,  
  "commercial_acceptance": 1, "sentiment": "pos", "ai_mention": 1, "spam": 0,  
  "aspects": [{"cat": "APPEARANCE", "pol": "neg"}, {"cat": "PERSONA", "pol": "pos"},  
               {"cat": "PRODUCT_BRAND", "pol": "neu"}]}
```

Add 8–12 few-shot examples taken from the development set, balanced across constructs and including hard negatives. Keep the exact prompt, model id, date, temperature and pre-processing rules in the paper's appendix.

Appendix B — Model specifications (R)

```

library(mgcv); library(segmented)

# Trajectories: comment i in post j of persona k
m_uv <- bam(uv ~ s(week, k=12) + s(week, by=vi, k=8, m=1)
           + sponsored + ai_disclosure + content_type + is_reply
           + visual_realism + coverage_ratio
           + s(post_id, bs="re") + s(vi, bs="re"),
           family=binomial, data=d, method="fREML", discrete=TRUE)
m_psi <- update(m_uv, psi ~ .); m_cr <- update(m_uv, cr ~ .)

# Curves at reference covariates, random effects excluded
grid <- expand.grid(week=seq(0, max(d$week), by=.5), sponsored=0, ai_disclosure=1,
                   content_type="photo", is_reply=0,
                   visual_realism=median(d$visual_realism),
                   coverage_ratio=1, vi=levels(d$vi)[1], post_id=levels(d$post_id)
                   [1])
R_hat <- predict(m_uv, grid, type="response", exclude=c("s(post_id)","s(vi)"),
                se.fit=TRUE)
S_hat <- predict(m_psi, grid, type="response", exclude=c("s(post_id)","s(vi)"),
                se.fit=TRUE)
D_hat <- S_hat$fit - R_hat$fit

# tau*: earliest week with lower 95% bound of D > 0 for >= 3 consecutive weeks.
# Cluster bootstrap: resample post_id with replacement (B = 500-1000), refit,
# recompute tau*.

# Rates of change: gratia::derivatives(m_uv, "s(week)")

# Phase structure (weekly series)
fit0 <- glm(cbind(n_uv, n - n_uv) ~ week, family=binomial, data=weekly)
pscore.test(fit0); fit1 <- segmented(fit0, seg.Z=~week, npsi=1); fit2 <-
segmented(fit0, seg.Z=~week, npsi=2)
AIC(fit0, fit1, fit2)

# Commercial receptivity (weekly):  $C_t \sim D_{t-1} + C_{t-1} + \text{sponsored\_share}_t + \text{product\_share}_t + \log(\text{volume}_t)$ 
# quasi-binomial GLM; Newey-West SE; sensitivity: lag 2, 2-week bins, drop W33

```

Appendix C — Analysis table schema (one row per comment)

FIELD	MEANING	SOURCE
<code>comment_id</code> , <code>parent_comment_id</code> , <code>is_reply</code>	identity and thread structure	crawler
<code>post_id</code> , <code>vi_id</code> , <code>content_type</code> , <code>caption</code>	post-level context	crawler / seed
<code>post_date</code> , <code>comment_time</code> , <code>time_precision</code> , <code>week_since_launch</code> , <code>bin2w</code>	time index	derived
<code>author_hash</code> , <code>author_n_comments</code>	repeat-commenter analysis	derived
<code>sponsored</code> , <code>product_category</code> , <code>ai_disclosure</code> , <code>visual_realism</code> , <code>social_content</code>	post-level coding (controls)	manual / LLM on caption
<code>ui_comment_count</code> , <code>n_collected</code> , <code>coverage_ratio</code>	collection coverage	crawler
<code>uv_resistance</code> , <code>ai_skepticism</code> , <code>parasocial_orientation</code> , <code>commercial_acceptance</code> , <code>sentiment</code> , <code>ai_mention</code> , <code>spam</code> , <code>aspects</code>	construct coding	LLM (+ human gold columns <code>*_gold</code>)
<code>lang_flag</code> , <code>empty_text</code> , <code>artefact</code>	cleaning flags	derived

11. References (verified)

Every entry below was checked against Crossref / CRAN / ACL Anthology / arXiv on 2026-08-21.

Grouped by where each belongs in the paper — marketing literature first. [§] = which section of this playbook it backs.

A. Virtual influencer marketing [§3, §8]

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